$$\frac{E\omega\left(e^{-\frac{s\pi}{\omega}}+1\right)}{\left(1+e^{-\frac{s\pi}{\omega}}\right)\left(1-e^{-\frac{s\pi}{\omega}}\right)(s^2+\omega^2)} = \frac{1-e^{-\frac{2\pi s}{\omega}}}{1-\left(e^{-\frac{\pi s}{\omega}}\right)^2}$$

$$f(t) = \begin{cases} k & 0<t<a \\ -k & a<t<2a \end{cases}, \quad f(t+2a) = f(t)$$

Given: $T = 2a$.

Wkt

$$L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)\, dt.$$

$$= \frac{1}{1-e^{-s(2a)}} \int_0^{2a} e^{-st} f(t)\, dt.$$

$$= \frac{1}{1-e^{-2as}} \left[\int_0^a e^{-st} k\, dt + \int_a^{2a} e^{-st}(-k)\, dt \right]$$

$$= \frac{k}{1-e^{-2as}} \left[\left[\frac{e^{-st}}{-s}\right]_0^a - \left[\frac{e^{-st}}{-s}\right]_a^{2a} \right]$$

$$= \frac{k}{(1-e^{-2as})(-s)} \left[\left(e^{-sa}-1\right) - \left(e^{-2as}-e^{-as}\right) \right]$$

$$= \frac{k}{(1-e^{-2as})(-s)} \left[e^{-sa} - 1 - e^{-2as} + e^{-as} \right]$$

$$= \frac{k}{(1-e^{-2as})(-s)} \left[-1 - e^{-2as} + 2e^{-as} \right]$$

$$= \frac{k(-1)}{(1-e^{-2as})(+s)} \left[1 + (e^{-as})^2 - 2e^{-as} \right]$$

$$= \frac{k}{(1-e^{-2as})(s)} \left[(1-e^{-as})^2 \right]$$

$$= \frac{k}{s(1+e^{-as})(1-e^{-as})} \left[(1-e^{-as})^2 \right]$$

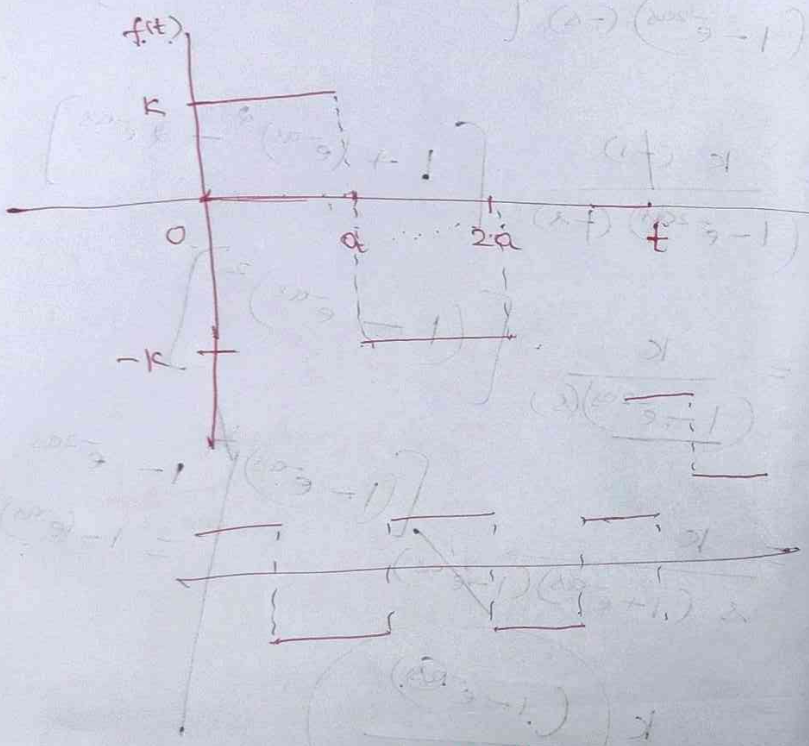
$$1-e^{-2as} = 1-(e^{-as})^2$$

$$= \frac{k}{s} \frac{(1-e^{-as})}{1+e^{-as}}$$

$$= \frac{k}{s} \tanh\frac{as}{2}$$

$$\tanh ax = \frac{\sinh ax}{\cosh ax} = \frac{(e^{ax}-e^{-ax})/2}{(e^{ax}+e^{-ax})/2} = \frac{e^{ax}(1-e^{-2ax})}{e^{ax}(1+e^{-2ax})}$$

$$f(t) = \begin{cases} k & 0 < t < a \\ -k & a < t < 2a \end{cases}$$



$$f(t) = \begin{cases} t & 0 < t < a \\ 2a - t & a < t < 2a \end{cases}$$

Given $T = 2a$

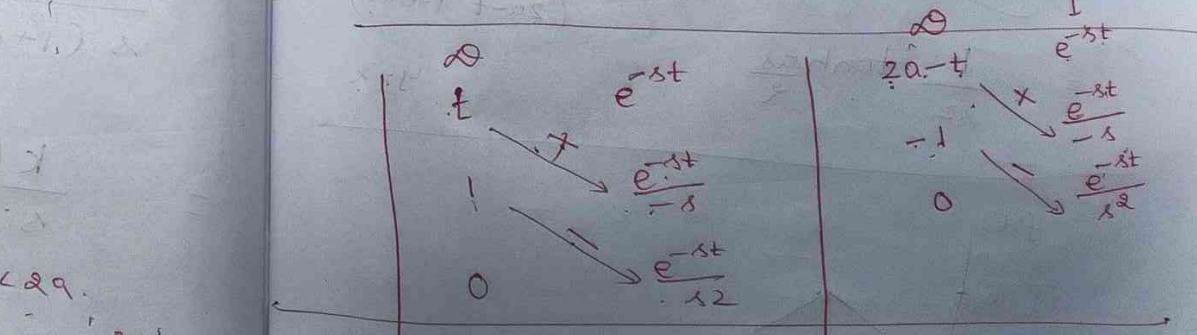
Wkt: $L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t)\, dt$

$$= \frac{1}{1 - e^{-s(2a)}} \int_0^{2a} e^{-st} f(t)\, dt$$

$$= \frac{1}{1 - e^{-2as}} \left[\int_0^a e^{-st} \cdot t \, dt + \int_a^{2a} e^{-st}(2a - t)\, dt \right]$$

$$= \frac{1}{1 - e^{-2as}} \left\{ \left[-t\frac{e^{-st}}{s} - \frac{e^{-st}}{s^2} \right]_0^a + \left[-(2a - t)\frac{e^{-st}}{s} + \frac{e^{-st}}{s^2} \right]_a^{2a} \right\}$$

$$= \frac{1}{1 - e^{-2as}} \left\{ \left[\left(-\frac{a e^{-sa}}{s} - \frac{e^{-sa}}{s^2} \right) - \left(0 - \frac{1}{s^2} \right) \right] + \left[\left(0 + \frac{e^{-2as}}{s^2} \right) - \left(-\frac{a e^{-sa}}{s} + \frac{e^{-sa}}{s^2} \right) \right] \right\}$$



$$= \frac{1}{1 - e^{-2as}} \left[-\frac{a e^{-sa}}{s} - \frac{e^{-sa}}{s^2} + \frac{1}{s^2} + \frac{e^{-2as}}{s^2} + \frac{a e^{-sa}}{s} + \frac{e^{-sa}}{s^2} \right]$$

$$= \frac{1}{1-e^{-2as}(s^2)} \left[1 + e^{-2as} - 2e^{-as} \right]$$

$$= \frac{1}{(1-e^{-2as})s^2} \left[1 + (e^{-as})^2 - 2e^{-as} \right]$$

$$= \frac{1}{(1-e^{-2as})(s^2)} \left(1 - e^{-as} \right)^2$$

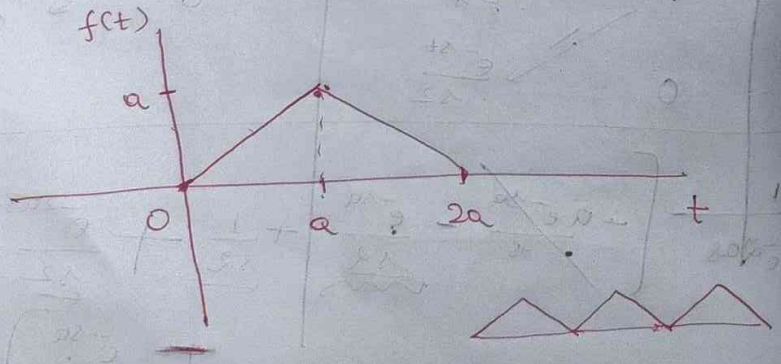
$$= \frac{1}{(1+e^{-as})(1-e^{-as})s^2} (1-e^{-as})^2$$

$$= \frac{1}{s^2} \cdot \frac{1-e^{-as}}{1+e^{-as}}$$

$$= \frac{1}{s^2} \tanh \frac{as}{2}$$

$$f(t) = \begin{cases} t & (0, a) \\ 2a - t & (a, 2a) \end{cases}$$

$y = x$



Unit Impulse.

① $\delta(t-a) = \lim_{\epsilon \to 0} \delta_\epsilon(t-a)$

$$L\{f(t)\delta(t-a)\} = e^{-as} f(a)$$

Note: $L\{\delta(t-a)\} = e^{-as}$

$L\{\delta(t)\} = 1$

$L\{f(t-a)\,u(t-a)\} = e^{-as} F(s)$

$L\{f(t)\delta(t-a)\} = e^{-as} f(a)$

① $L\{\sin\sqrt{3}t\,\delta(t-3)\} = e^{-3s} \sin 3\sqrt{3}$

② $L\{e^{-4t}\,\delta(t-e)\} = e^{-2s}\, e^{-8}$

③ $L\{t^5\,\delta(t-5)\} = e^{-5s}\,(5^5)$

$$\frac{(s+3)-(s+1)}{(s-3)(s+3)} = \frac{1}{2}\left(\frac{6}{s^2-9}\right)$$

Inverse L.T

$L\{k\} = \frac{k}{s}$

$L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$

$L\{t^n\} = \frac{n!}{s^{n+1}}$

$L\{1\} = \frac{1}{s}$

$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$

$L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{1}{a}\sin at$

$L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$

$L^{-1}\left\{\frac{1}{s^2-a^2}\right\} = \frac{1}{a}\sinh at$

$L^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$

$L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{\Gamma(n+1)}$

Eg: $L^{-1}\left\{\frac{1}{s^{10}}\right\} = \frac{t^9}{9!}$

$L^{-1}\left\{\frac{1}{s^{7/2}}\right\} = \frac{t^{5/2}}{\Gamma(5/2+1)}$

$L^{-1}\left\{\frac{1}{s^6}\right\} = \frac{t^5}{5!}$

① $F(s) = \frac{1}{s} + \frac{1}{s-5} + \frac{1}{s+6} + \frac{1}{s^8} + \frac{1}{s^{7/2}} + \frac{1}{s^2+(\sqrt{3})^2} + \frac{s}{s^2+(\sqrt{5})^2}$

$= 1 + e^{5t} + e^{-6t} + \frac{t^7}{7!} + \frac{t^{5/2}}{\Gamma(5/2+1)} + \frac{1}{\sqrt{3}}\sin\sqrt{3}t + \cos\sqrt{5}t$ ...

$\Gamma(5/2+1)$

$\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}$

Shifting

$\frac{s+3}{(s+3)^2+25} = e^{-3t}L^{-1}\left\{\frac{s}{s^2+5^2}\right\}$

$= e^{-3t}\cos 5t$

Completing square

s^2+3s+5

$\left(s+\frac{3}{2}\right)^2 + 5 - \frac{9}{4}$

$\left(s+3/2\right)^2 + \frac{11}{4} = \left(s+3/2\right)^2 + \left(\frac{\sqrt{11}}{2}\right)^2$

① $\frac{s-2}{s^2-4s+8}$

Dr s^2-4s+8

$\left[s-\frac{1}{2}(4)\right]^2+8-4$

Dr: $(s-2)^2+2^2$

$L^{-1}\left\{\frac{s-2}{(s-2)^2+2^2}\right\}$

$s-2$ to s

$= e^{2t}L^{-1}\left\{\frac{s}{s^2+2^2}\right\}$

$= e^{2t}\cos 2t$

② $\frac{1}{s^2+6s+18}$

Dr: $s^2+6s+18$

$\left(s+\frac{1}{2}(6)\right)^2+18-9$

$(s+3)^2+3^2$

$L^{-1}\{\ \}$

$= \frac{1}{(s+3)^2+3^2}$

$s+3$ to s

$= e^{-3t}L^{-1}\left\{\frac{1}{s^2+3^2}\right\}$

$= e^{-3t}\frac{1}{3}\sin 3t$

$\frac{7s+4}{4s^2+4s+9}$

Dr: $4s^2+4s+9$

$4\left[s^2+s+\frac{9}{4}\right]$

$= 4\left[\left[s+\frac{1}{2}(1)\right]^2+\frac{9}{4}-\frac{1}{4}\right]$

$= 4\left[(s+1/2)^2+(\sqrt{2})^2\right]$

Nr: $7s+4$

$= 7\left(s+\frac{1}{2}-\frac{1}{2}\right)+4$

$= 7(s+1/2)-\frac{7}{2}+4$

$= 7(s+1/2)+\frac{1}{2}$

$\therefore \frac{7s+4}{4s^2+4s+9} = \frac{7(s+1/2)+\frac{1}{2}}{4\left[(s+\frac{1}{2})^2+(\sqrt{2})^2\right]}$

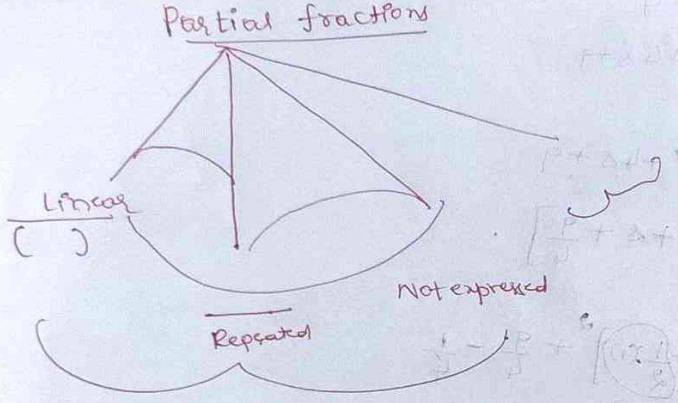
$= \frac{1}{4}\left\{\frac{7(s+1/2)}{(s+1/2)^2+(\sqrt{2})^2}+\frac{1/2}{(s+1/2)^2+(\sqrt{2})^2}\right\}$

$= \frac{1}{4}\left\{e^{-\frac{1}{2}t}L^{-1}\left\{\frac{7s}{s^2+(\sqrt{2})^2}\right\}+\frac{1}{2}e^{-1/2t}L^{-1}\left\{\frac{1/\sqrt{2}}{s^2+(\sqrt{2})^2}\right\}\right\}$

$s+1/2$ to s

$\frac{e^{-1/2t}}{4}\left(7\cos\sqrt{2}t+\frac{1}{2}\frac{1}{\sqrt{2}}\sin\sqrt{2}t\right)$

Partial fractions



① $\frac{s+k}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b}$

② $\frac{s+4}{(s+k_1)^2(s+k_2)} = \frac{A}{s+k_1} + \frac{B}{(s+k_1)^2} + \frac{C}{(s+k_2)}$

③ $\frac{s+a}{(s+k_1)(s^2+k_2)} = \frac{A}{s+k_1} + \frac{Bs+C}{s^2+k}$

$\frac{s^2-10s+13}{(s+1)(s^2-5s+6)}$

$s=3$
$s=2$

$\frac{s^2-10s+13}{(s+1)(s-3)(s-2)} = \frac{A}{s+1} + \frac{B}{s-3} + \frac{C}{s-2}$

① ② ③ ② ①

Compare

$\frac{s^2-10s+13}{(s+1)(s-3)(s-2)} = \frac{A}{s+1} \Rightarrow A = \frac{24}{12} = 2$ $\quad A=2$

$s=-1$

Consider

$\frac{s^2-10s+13}{(s+1)(s-3)(s-2)} = \frac{B}{s-3} \Rightarrow B = \frac{9-30+13}{4}$

$B=-2$

$s=3$

Consider

$\frac{s^2-10s+13}{(s+1)(s-3)(s-2)} = \frac{C}{s-2} \Rightarrow \frac{4-20+13}{-3} = C$

$C=1$

$s=2$

① $\Rightarrow$

$F(s) = \frac{2}{s+1} + \frac{2}{s-3} + \frac{1}{s-2}$

$= 2e^{-t} - 2e^{3t} + e^{2t}$

$$\frac{1}{(s+1)(s+7)^2} = \frac{A}{s+1} + \frac{B}{s+7} + \frac{C}{(s+7)^2}$$

① ② ① ② → ①

$$\times (s+1)(s+7)^2$$

$$1 = A(s+7)^2 + B(s+1)(s+7) + C(s+1) \quad \text{→ ②}$$

Consider

$$\left.\frac{1}{(s+1)(s+7)^2}\right|_{s=-1} = \frac{A}{s+7} \Rightarrow \boxed{\frac{1}{36} = A}$$

$$\left.\frac{1}{(s+1)(s+7)^2}\right|_{s=-7} = \frac{C}{(s+7)^2} \Rightarrow \boxed{-\frac{1}{6} = C}$$

From ② Compare the coefficient of s^2

$$0 = A + B$$

$$\boxed{B = -\frac{1}{36}}$$

① ⇒

$$F(s) = \frac{1}{36}\left[\frac{1}{s+1}\right] - \frac{1}{36}\left[\frac{1}{s+7}\right] - \frac{1}{6}\left[\frac{1}{(s+7)^2}\right]$$

$s+7$ to s

$$= \frac{1}{36}e^{-t} - \frac{1}{36}e^{-7t} - \frac{1}{6}e^{-7t}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$\text{——— ||} \quad - \frac{1}{6}e^{-7t}\, t$$